# Gautham Ramkumar

Ashburn, VA | +1 (857) 565-5374 | ramkumar.g@northeastern.edu
linkedin.com/in/rgautham20 | github.com/gautham-ramkumar | gautham-ramkumar.github.io
Authorized to work in the US for 3 years with no sponsorship required

## KEY ACHIEVEMENTS

- Placed 2nd at Revolute Hackathon, shipping a 3-model voice-to-action manipulation pipeline in one weekend

## EDUCATION

**Northeastern University, Boston, MA** May 2026
*Master of Science in Robotics (Concentration: ECE)* *GPA: 3.636/4.0*

**Vellore Institute of Technology, India** May 2024
*Bachelor of Technology in Electronics and Communications Engineering* *GPA: 3.48/4.0*

## TECHNICAL SKILLS

| | |
|---|---|
| **Languages** | C++17, Python |
| **Control & Dynamics** | Hybrid Zero Dynamics, Feedback Linearization, Trajectory Optimization, Legged Locomotion |
| **Autonomy** | Visual SLAM (RTAB-Map), Visual-Inertial Odometry, EKF (GNSS/IMU), GTSAM/iSAM2 |
| **Hardware** | NVIDIA Jetson Orin Nano, LiDAR, IMU, GNSS, SocketCAN, 6-DoF Arm |
| **Software Practices** | ROS2, ROS1 (Kalibr), Docker, Linux, Git, PyTest, GitHub Actions CI/CD, MuJoCo |

## PROFESSIONAL EXPERIENCE

**Chicago Education Advocacy Corporation** Aug 2026 – Present
*Robotics AI Engineer, Chicago, IL (Remote)*

- Grew automated test coverage from 44 to 145 tests across a two-job CI pipeline
- Reproduced a 70% SmolVLA task-success baseline in MuJoCo on an 8 GB GPU with pinned dependencies
- Built a dataset verifier that catches 3 silent defects (zeroed chunks, wrong state convention, no unnormalization)
- Built Backstop, a VLA failure guard that stops failing runs 0.67 s sooner than a timer at 4.3% false stops

**SmartInternz** May 2023 – Jul 2023
*Internet of Things Intern, Chennai, India*

- Maintained 1 cm distance accuracy in a real-time ultrasonic sensing pipeline for automated access control

## PROJECTS

**YesChef: Voice-Controlled Robotic Sandwich Assembly** Aug 2026
*Python, LeRobot, Jetson Orin Nano, SocketCAN | Demo*

- Added bounded retry logic around CAN control-ack timeouts, stopping crashes on the 6-DoF arm's control loop
- Fixed a leader-arm unwrap fault, rescaling the gripper so 270° of follower range spans 90° of leader travel
- Chained Whisper, Qwen2.5-3B, and Qwen2.5-VL-3B into trajectory execution on a 6-DoF arm, on-device

**Hybrid Zero Dynamics Bipedal Gait Control** 2026
*MATLAB, Feedback Linearization, fmincon | Code*

- Imposed 4th-order Bézier virtual constraints to reduce a 3-link planar biped onto a 2-DOF Hybrid Zero Dynamics manifold, optimized with fmincon
- Synthesized input-output feedback linearization and validated 1.73 m/s walking at 1.1584 MCOT, Poincaré-stable over 15 steps

**Indoor 3D Mapping using RTAB-Map SLAM** Sep 2025 – Oct 2025
*ROS2, C++, ZED Mini | Demo*

- Achieved 90% ground-truth alignment deploying visual SLAM on ZED Mini stereo hardware in ROS2
- Mapped 250 m of low-texture, GPS-denied tunnel; fixed handheld tracking loss with a rolling-chair mount

**Camera-LiDAR-IMU Extrinsic and Temporal Calibration** Oct 2025 – Dec 2025
*C++17, Python, ROS2, GTSAM, PyTest | Code*

- Rejected an unstable 150ms camera-LiDAR offset; 748 timestamp pairs gave -19.5ms within 1.5ms over 15 windows
- Tripled usable LiDAR point density per scan via a KD-tree dynamic-point filter, no segmentation network